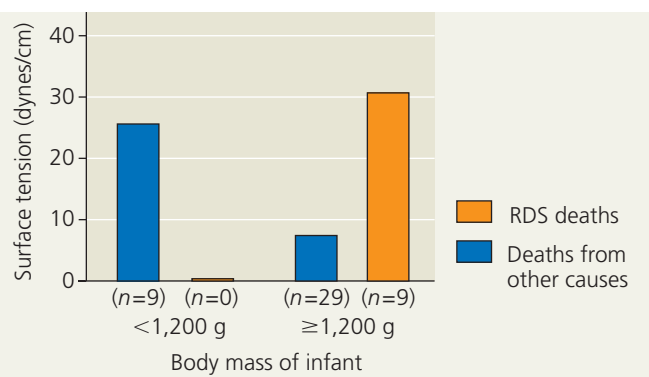


▼ Figure 42.25 Inquiry

What causes respiratory distress syndrome?

Experiment Mary Ellen Avery, a research fellow at Harvard University, hypothesized that respiratory distress syndrome (RDS) in preterm infants is caused by the lack of a substance that reduces surface tension in the alveoli. To test this idea, she obtained autopsy samples of lungs from infants who had died of RDS or from other causes. She extracted material from each sample and let it form a film on water. Avery then measured the tension (in dynes per centimeter) across the water surface for each sample.

Results Avery noted a pattern when she separated out samples from infants with body mass less than 1,200 g, the average mass of a fetus at 29–30 weeks of pregnancy.



Data from M. E. Avery and J. Mead, Surface properties in relation to atelectasis and hyaline membrane disease, *American Journal of Diseases of Children* 97:517–523 (1959).

Conclusion For infants with a body mass of 1,200 g or greater, samples from those who had died of RDS exhibited much higher surface tension than samples from those who had died from other causes. Avery inferred that infants' lungs normally contain a surface-tension-reducing substance (now called surfactant) and that RDS occurs when this substance is lacking. The results from infants with a body mass less than 1,200 g were similar to those from infants who had died from RDS, suggesting that surfactant is not normally produced until a fetus reaches this size.

WHAT IF? If the researchers had measured the amount of surfactant in lung samples from the infants, what relationship would you expect between the amount of surfactant and infant body mass?

Having surveyed the route that air follows when we breathe, we'll turn next to the process of breathing itself.

CONCEPT CHECK 42.5

1. Why is an internal location for gas exchange tissues advantageous for terrestrial animals?
2. After a heavy rain, earthworms come to the surface. How would you explain this behavior in terms of an earthworm's requirements for gas exchange?
3. **MAKE CONNECTIONS** Describe similarities in the counter-current exchange that facilitates respiration in fish and thermoregulation in geese (see Concept 40.3).

For suggested answers, see Appendix A.

CONCEPT 42.6

Breathing ventilates the lungs

Like fishes, terrestrial vertebrates rely on ventilation to maintain high O₂ and low CO₂ concentrations at the gas exchange surface. The process that ventilates lungs is **breathing**, the alternating inhalation and exhalation of air. A variety of mechanisms for moving air in and out of lungs have evolved, as we will see by considering breathing in amphibians, birds, and mammals.

How an Amphibian Breathes

An amphibian such as a frog ventilates its lungs by **positive pressure breathing**, inflating the lungs with forced air-flow. Inhalation begins when muscles lower the floor of an amphibian's oral cavity, drawing in air through its nostrils. Next, with the nostrils and mouth closed, the floor of the oral cavity rises, forcing air down the trachea. Exhalation follows as air is expelled by the elastic recoil of the lungs and by compression of the muscular body wall. When male frogs puff themselves up in aggressive or courtship displays, they disrupt this breathing cycle, taking in air several times without allowing any release.

How a Bird Breathes

When a bird breathes, it passes air over the gas exchange surface in only one direction. Air sacs situated on either side of the lungs act as bellows that direct air flow through the lungs. Within the lungs, tiny channels called *parabronchi* serve as the sites of gas exchange. Passage of air through the entire system—air sacs and lungs—requires two cycles of inhalation and exhalation (**Figure 42.26**).

Ventilation in birds is highly efficient. One reason is that birds pass air over the gas exchange surface in only one direction during breathing. In addition, incoming fresh air does not mix with air that has already carried out gas exchange, maximizing the partial pressure difference with blood flowing through the lungs.

time killed 10,000 infants annually in the United States (**Figure 42.25**). RDS is most common among infants born before 32 weeks of pregnancy. (The average full-term human pregnancy is 38 weeks.) Later studies revealed that surfactant typically appears in the lungs after 33 weeks of gestation. Artificial surfactants are now used to treat early preterm infants, and treated babies with a body mass over 900 g (2 pounds) at birth usually survive without long-term health problems. For her contributions, Avery received the National Medal of Science.